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Multiphase flow separation is a critical technological challenge for the human exploration of space and planetary bodies. While droplet and bubble segregation is a spontaneously occurring phenomenon in most terrestrial situations, the absence of the gravity-induced separatory action in a microgravity environment often results in situations where two disparate phases have no distinct inclination to separate from one another. This can present many problems with respect to spacecraft and space station operations as proper control and sequestration of the individual phases is often essential to the successful functionality of many system-critical components.
Passive cyclonic separators are a promising technology that potentially could fulfill the separation needs of many of these systems. While considerable research has been performed on this technology, the limitations of terrestrial testing, aircraft testing, and modeling necessitates long-term experimentation to raise these devices to a sufficient technology readiness level (TRL) to permit use within spacecraft systems. It is a lack of basic engineering and scientific knowledge

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that currently prevents these systems from being employed. At a basic engineering level, there is a distinct lack of data regarding both the steady and unsteady operational performance of vortex separators.

Issues such as where the designs can be expected to work and how to go about optimizing designs are unresolved. Accurate prediction of these phenomena using CFD is currently not tenable due to the dearth of available data and incomplete modeling techniques. Beyond engineering, there is also a lack of knowledge regarding the fundamental multiphase fluid physics phenomena that occur within passive cyclonic separators. Issues such as gas core adjustment to injection flow rates, bubble dynamics, vortex flow interaction with capillary forces at the gas core, etc. are all unresolved. Inasmuch, this research effort seeks to resolve these issues. The main objectives are to:

1. Quantitatively assess and delimit the range of operability of a prototype passive cyclonic separator including the transient and steady-state instability behavior of the device with respect to both device and system-level conduct.

2. Elucidate and quantify the multiphase fluids physics phenomena governing the performance of microgravity two-phase separators.

3. Validate and verify the multiphase CFD modeling approaches that have been developed to simulate the complicated two-phase transport processes associated with phase separation in microgravity.

4. Validate and verify a hydraulic system code for the simulation of two-phase flow in microgravity.

To achieve these objectives, combined experimental, numerical, and phenomenological modeling analyses are being performed to quantitatively describe the performance and response of systems containing separators. Specifically, the work entails the creation of multiphase flow system that is able to provide metered quantities of gas and liquid into a test separator. The separator and the associated flow loop had been operated under a variety of specially tailored conditions, including specific combinations of parameters to examine flow features of interest including the steady-state performance margins of the separator, the transient response of the separator, vapor core instability and misting, system startup/shutdown, and system level performance and instabilities. The products of this research have included the flight experiment (including two-phase flow loop and passive cyclonic separator), validated and verified CFD modeling techniques, a microgravity hydraulic system code, and archival data capturing the important fluid physics associated with microgravity separator operations.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

From a microgravity/scientific standpoint, the experimentation, modeling, and analytical work of this research will serve to:
1. Provide elucidation and understanding of the above-mentioned two-phase flow phenomena associated with the operation of separators in multiphase systems

2. Provide details of the columnar stability of the gas core

3. Provide details of the time constants associated with the response of passive cyclonic separators

4. Validate and verify a coupled CFD and hydraulic system code for simulations of two-phase flow in microgravity

The resulting insight into phase separation, distribution, and control will help to afford NASA designers the latitude to take greater advantage of the benefits offered by the use of these multiphase systems in spacecraft applications. The experimental results will also facilitate further new and original design modifications to improve the performance of microgravity phase separation devices. Validation and verification of the numerical results are expected to permit the use of CFD as a reliable design tool to further assist in the creation and analysis of new separator designs, while the results from the system code are expected to assist in the creation and analysis of general multiphase systems in microgravity.

Overall, the research will help contribute to the fundamental scientific understanding of two-phase fluid behavior, both in microgravity and 1-g environments.

Numerical Modeling
The computational studies of the fluid physics within the separation device conducted during the past year of the grant have been continued. Two hybrid multiphase CFD techniques were developed, using OpenFOAM that could model the unique flow features in microgravity cyclonic separators. The Large Eddy Simulation (LES) turbulence model was implemented to capture the self-induced unsteadiness of the gas core behaviors. This multiscale approach could handle the disperse phase while also capturing the relevant capillarity features at gas core. The CFD techniques were then used to simulate the separator performance under the conditions of steady injection of both single-phase and multiphase flow and to gain insight of the separation process.

The hybrid multiphase model which combined the Eulerian multi-fluid approach with VOF (volume of fluid) sharp interface capturing has been continued to improve during the past year. The coupling between these two approaches is based on dynamic, local switching of interface sharpening algorithm. Extension of this hybrid model into a three-dimensional simulation for our separator was made to gain detailed insight into the internal flow physics and to assist the design of an efficient separator. This hybrid multiphase approach coupled with LES approach presented a very realistic simulation which captured the anisotropic characteristic of swirl flow and the self-induced unsteadiness associate with gas core in the cyclone separator. A gamma cutoff value of 0.4 for criterion of dynamic interface sharpening seemed to given a reasonable prediction for gas core diameter. Comparisons of the gas core and pressure variation were compared to experimental results, and satisfactory agreement was found.

For bubble dynamics, the combined VOF method with a two-way coupled Lagragian bubble tracking approach was developed and conducted to predict bubble trajectories as well as gas core behavior. The continuous phase was simulated using the VOF approach to simulate large scale gas core and to capture the capillarity at the interface, while the discrete bubbles were computed in a Lagrangian stochastic model. As the experiments do not provide particularly though comparative data, the general performance of the separator was to be delimited for logical consistency. The effects of bubble size and inlet liquid flow rate on separator efficiency were examined. The smallest bubble size that is

Task Progress:	successfully separated by the separator for an assumed flow condition is able to be predicted by this hybrid model. The effects of turbulence dispersion were also included in the bubble trajectory simulation. It is important to address the influence of turbulence on small bubbles which are more responsive to velocity fluctuations. Moreover, the presence of the baffle plate has been shown that it reversed the axial flow of gas bubbles that has been directed to the center of the separator so that the residence time of the gas bubbles within the swirling flow region increased. The increased residence time of gas bubbles assisted the completion of the separation process. Analytical Modeling The analytical approach has continued to provide insight into device design. The control-volume-based analytical approach (initially developed in previous years) has been extended to account for capillary effects. The modifications required the addition of terms to account for pressure gradient induced by surface tension force at the interface. Surface tension effects could become important in microgravity, which adversely affects the separator performance. The collapse of gas core due to surface tension is investigated using control volume analysis to determine the condition for the gas core collapse. The stability map of the gas core was created. The critical Weber numbers predicted by control-volume analysis agree reasonably well with CFD results. Experimentation The characteristic bubble diameter does indeed play a significant role with respect to the successful functionality of the separation action. For the study of bubble size distribution, the experimental techniques had been developed to fully characterize the injected bubble size distribution. Experiments will be conducted to study effects of various parameters on the bubble size distributions, including the flow rate (or the nozzle Reynolds number), volumetric quality, upstream two-phase flow structure, and nozzle size and length. The preliminary result showed that bubble size distribution at injection point can be characterized by a log-normal distribution in a range of 0.1 to 4 mm with a representative peak size of approximate 0.3 mm. Supporting Work for Flight Hardware Design A breadboard model of the proposed experiment is being designed and built at NASA Glenn Research Center (GRC). We have collaborated with NASA/GRC engineers to prepare and design the breadboard experiment system in past year. The void fraction sensors made by Tech4 are being calibrated in past year. A total of five sensors are used: three for 1” pipe, one for ¾” pipe, and one for ½” pipe. They are calibrated using the flow loop with three quick closing valves (QCV), which has been described in our earlier report. Also in our earlier tests, the sensor outputs were found to be very much affected by the edge effects. Consequently, even though each sensor is normalized before each test against pure air (capacitance value = 0) and pure water (capacitance value = 1), the normalized capacitance output with two-phase flow is always larger than unity due to the edge effects. In any case we proceeded to calibrate the sensors despite this problem. One consequence of this problem is that the capacitance output may depend on the flow regime of the two-phase flow. Earlier we have reported the calibration curves obtained in the bubbly flow regime. In the present report the calibration is done in the slug flow regime.
Bibliography Type:	Description: (Last Updated: 03/06/2024)
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Articles in Peer-reviewed Journals	Kharraz A, Kang MF, Kamotani Y. "Stability of gas core of cyclonic two-phase separator in microgravity." Interfacial Phenomena and Heat Transfer, Accepted as of June 2016. (SI: Two-Phase Systems for Space and Ground Applications) , Jun-2016